

Lumpy Adjustment: New Evidence and Aggregate Implications

Work in Progress

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Quick Summary

- Firms do not invest a little bit every year.
- Instead, they invest in *big, infrequent jumps* when they expand, modernize, or upgrade.
- These investment “bursts” are irregular — but they are *not random*.
- They often cluster across firms and can move the whole economy.

We study whether, and how, these investment bursts combine to drive business cycles.

- **Business cycle shock** → A sudden improvement or slowdown in the economy.
- **Extensive margin** → How many firms decide to invest this period.
- **Intensive margin** → How much each investing firm spends when it invests.
- **Impulse response** → How firms or the economy adjust over time after a shock hits.

- Firms adjust capital in *large, lumpy jumps* — not smooth increments.
- These jumps tend to **cluster**: many firms invest around the same time.
- When this clustering happens, **aggregate business cycle movements emerge**.
- We separate investment into “lumpy” vs. “non-lumpy” components.
- Only the **lumpy component** reacts strongly to economic shocks.
- Therefore, the **extensive margin** — how many firms invest — is the key micro foundation of the business cycle.

Context

- A well-established empirical fact: *firm-level investment is lumpy*. (Doms and Dunne (1998); Cooper and Haltiwanger (2005); Becker et al. (2006))
- **What is lumpiness?** Firms tend to adjust capital in *rare but very large spikes*, followed by long periods of near *inaction*.
- These sharp spikes occur when firms install new vintages of capital, reorganize production, or expand capacity—leading to *discontinuous and uneven* investment behavior.
- Lumpiness is observed across industries, firm sizes, and time periods, and is a foundational micro feature underlying modern heterogeneous-firm macro models.

Lumpiness at the Firm Level - Compustat U.S. Data 1981-2022

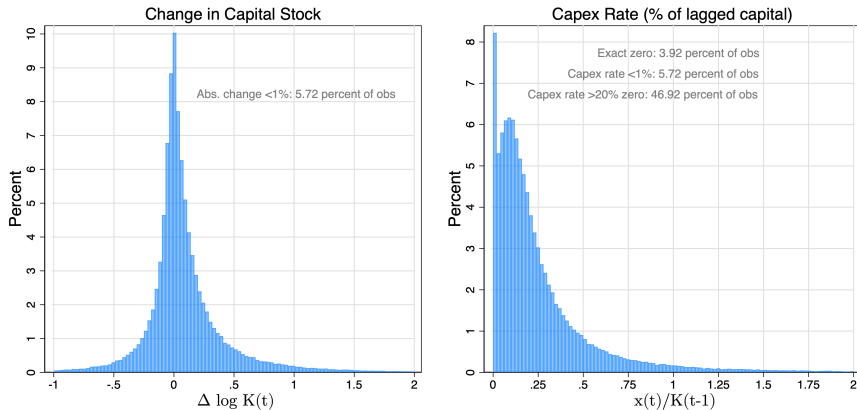


Figure 1: Lumpiness in U.S. Compustat Data of Public Firms

Agg Investment Rate = Lumpy + Nonlumpy Aggs

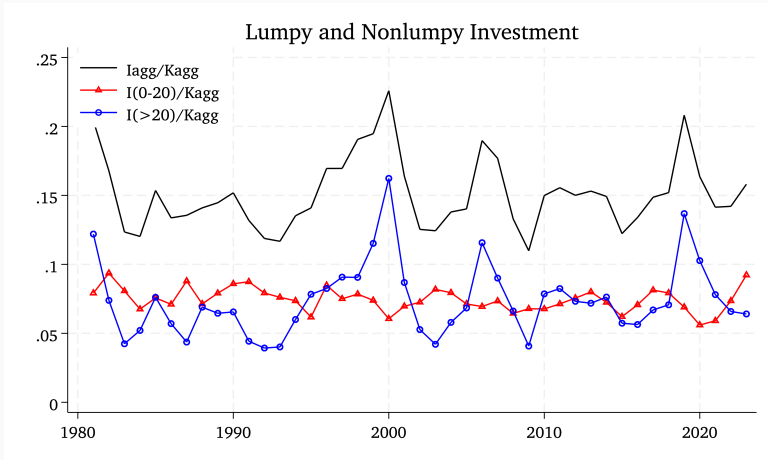


Figure 2: Investment Rate (I_{agg}/K_{agg}), Non Lumpy Investment Rate ($I(0 - 20)/K_{agg}$) and Lumpy Investment Rate ($I(> 20)/K_{agg}$)

- Agg investment rate and its “Lumpy” part seem to co-move.

Lumpy Adjustments = Intensive + Extensive Margin

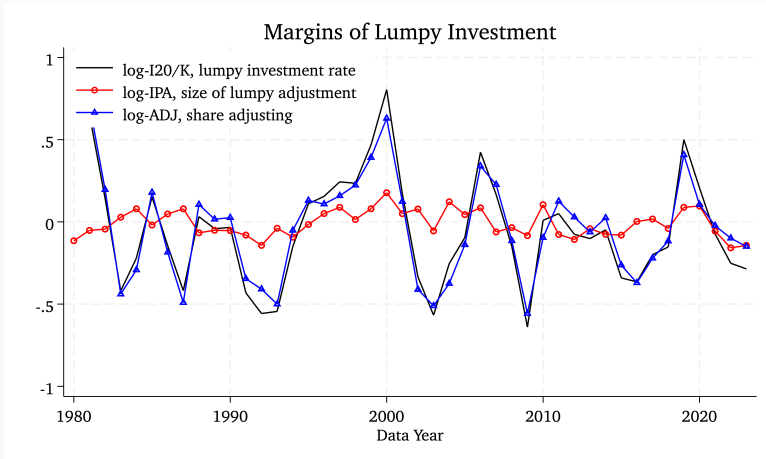


Figure 3: Lumpy Adjustments (I_{20}/K) = Intensive Adjustments (IPA) + Extensive Adjustments (ADJ)

- Extensive Margin - the share of firms adjusting lumpily seem to be more important than the Intensive Margin - the size of lumpy adjustment.

- Aggregate investment is not lumpy.
- The cross-sectional distribution of capital holdings is potentially important.
- **Does it matter for the aggregate?**
 - Thomas (JPE, 2002) and Khan and Thomas (Econometrica, 2008) show that distribution of capital is irrelevant for aggregate investment and output. GE effects: consumption smoothing affects interest rates and desired capital!
 - Bachmann, Caballero and Engel (AEJ: Macro, 2013) find agg. investment is conditionally heteroskedastic, state dependent response, can be rationalized with lumpy investment. Similar results by Winberry (AER, 2021).
- **Which margin is more important - Extensive vs Intensive?**
 - Guorio and Kashyap (2007) show that it is the number of firms making large investments (extensive margin) that determines bulk of aggregate variation rather than the average size of the spike (intensive margin).

Our Work

- Existing literature use General Equilibrium theoretical models.
- In contrast, we use a “novel” empirical approach that combines:
 - A Dynamic Factor Model (DFM) at the firm level, and
 - A Vector Auto-Regression (VAR) at the aggregate level
- Identify business cycle shocks and map their responses to aggregated/dis-aggregated firm outcomes.
- This framework is readily adaptable to other micro-macro stories.
- Our empirical evidence shows that:
 - **Lumpy adjustments significantly impact aggregate outcomes**
 - **Aggregate movements are driven by the extensive margin rather than the intensive margin**

Data

- U.S. firm (public) level data:
 - Compustat. Around 14,400 firms covering years 1981-2022.
 - Annual frequency
- Additionally, we use macro variables like GDP in the VAR.

Main Firm-Level Variables:

- Standard measurement following the literature (Ottonello and Winberry (2020)).
- **Capital (K_{it})**: Measured as PPE under the perpetual inventory method.
- **Investment (x_{it})**: Capital expenditure (capex) spending.
- **Firm Investment Rate ($\frac{x_{it}}{K_{it-1}}$)**: The ratio of current capex to lagged capital.
- **Aggregated Measures:**

$$I_{agg} = \sum_i x_{it}, \quad (1)$$

$$K_{agg} = \sum_i K_{it-1}. \quad (2)$$

Aggregated Lumpy Variables:

- **Lumpy Investment:** The aggregate capex of firms that make lumpy adjustments,

$$I_{20} = \sum_{i \in \mathcal{L}} x_{it}, \quad (3)$$

where $\mathcal{L}$ denotes the set of firms that make lumpy capital adjustments.

- **Lumpy Adjusters' Capital Stock:** The total lagged capital for these firms,

$$K_{20} = \sum_{i \in \mathcal{L}} K_{it-1}. \quad (4)$$

Extensive, Intensive margins of Lumpy Adjustment

- Following Guorio and Kashyap (2007), lumpy events can be decomposed into extensive and intensive margins

$$\log \frac{I_{20}}{K_{agg}} = \log \frac{I_{20}}{K_{20}} + \log \frac{K_{20}}{K_{agg}} \quad (5)$$

$$= \log\text{-}IPA + \log\text{-}ADJ \quad (6)$$

$$= \textit{size of adjustment} + \textit{share adjusting} \quad (7)$$

- IPA - investment per adjuster (Intensive Margin)
- ADJ - number of firms adjusting (Extensive Margin).

Econometric Model

- We implement a DFM-FAVAR approach to link firm-level dynamics with aggregate outcomes.
- Each firm-level variable is decomposed into:
 - A common component capturing cross-sectional co-movements, and
 - An idiosyncratic component.
- The baseline decomposition is given by:

$$x_{i,t} = \Lambda_{ij,t} C_t + \epsilon_{i,t}, \quad \epsilon_{i,t} \sim N(0, V_t) \quad (8)$$

- Here, $i = 1, \dots, N$ indexes firms, $j = 1, \dots, r$ indexes the common factors, and $t = 1, \dots, T$ indexes time; V_t is diagonal.

- **Time-Varying Loadings:**

- The factor loadings evolve over time, following a random walk:

$$\Lambda_{ij,t} = \Lambda_{ij,t-1} + \eta_t, \quad \eta_t \sim N(0, W_t) \quad (9)$$

- **Common Components Evolution:**

- The vector of common factors evolves according to a VAR process:

$$C_t = \beta C_{t-1} + u_t, \quad u_t \sim N(0, Q) \quad (10)$$

- Equations (1) and (2) jointly define the DFM-FAVAR system with time-varying loadings.

- **Defining Lumpy Adjustments:**

- Spikes in capital or labor adjustments are classified as lumpy.
- A capital adjustment is considered lumpy if the investment-to-capital ratio exceeds a threshold:

$$\frac{I_t}{K_{t-1}} > 0.2.$$

- A similar threshold is applied for labor adjustments.
- Threshold value consistent with Cooper and Haltiwanger (2006), Guorio and Kashyap (2007) and Görtz et al. (2023).

- **DFM Restrictions:**

- In practice, we impose either one-to-one restrictions ($\Lambda_{ij,t} = 1$) or exclusion restrictions ($\Lambda_{ij,t} = 0$) on the sequence of firm-level loadings.
- Additional restrictions address factor rotational and scale indeterminacies.
- *No restrictions needed when we use observed factors (firm level aggregates).*

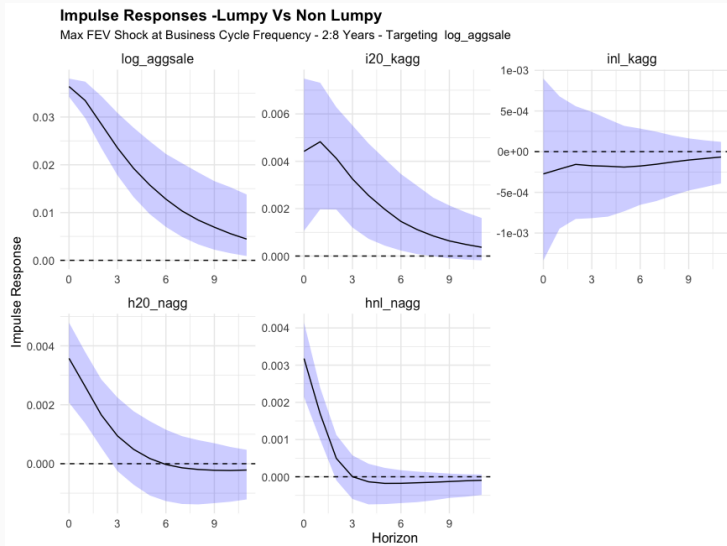
- We identify two aggregate shocks:
 1. Business Cycle Shock
 2. Financial Shock
- Both shocks are identified in an agnostic, purely statistical manner using the Max–Forecast Error Variance (Max-FEV) approach, following Uhlig (2003), Kurmann and Otrok (2013), and Angeletos et al. (2020).
- **Business Cycle Shock:** the structural innovation that maximizes the FEV of a chosen real activity indicator over a finite horizon.
Real Activity Target variables: (i) Aggregate Firm Sales, (ii) GDP.
- **Financial Shock:** the structural innovation that maximizes the FEV of a financial conditions indicator over a finite horizon.
Financial Conditions Target variable: Excess Bond Premium (Gilchrist and Zakrajšek, 2012).

- **Common Components Construction:**
 - **Aggregate Approach:** Use macro-level variables such as GDP along with aggregated firm level variables such as:
 - Firm Sales
 - Lumpy and Non lumpy Adjustments
 - Extensive and Intensive Adjustments
 - **PCA Approach:** Extract principal components from relevant firm-level variables to form factors (e.g., Sales Factor, TFP Factor, lumpy and non-lumpy factors).
- These components serve as the building blocks for our DFM-FAVAR system.

- Equations (1) and (2) define a linear Gaussian state-space model. We obtain Kalman smoothed estimates of the time-varying loadings, $\Lambda_{ij,t}$.
- Variance terms are estimated using an EWMA approach, following Koop and Korobilis (2014).
- The VAR in Equation (3) is estimated using standard VAR procedures.
- Conditional on known (or initialized) values of $\tilde{C}_t$ and estimated parameters $\lambda_{ij,t}$, β , V_t , W_t , and Q , we update the Kalman smoothed estimates for C_t .

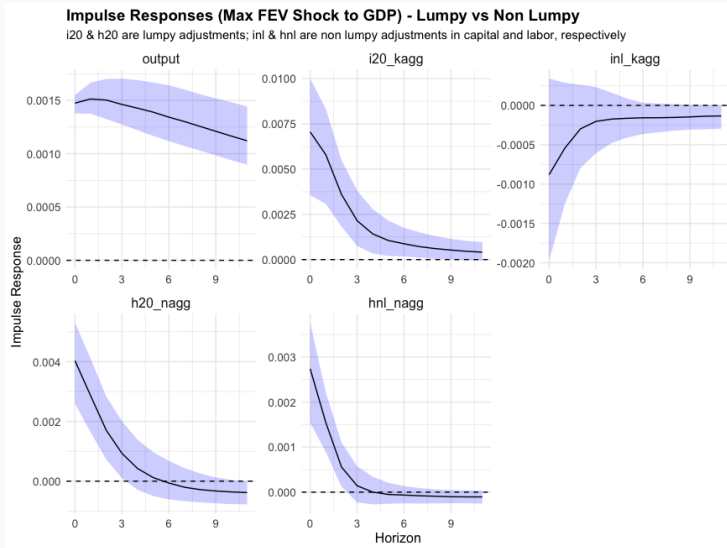
Results - Aggregate Dynamics

Impulse Responses - Lumpy vs Non Lumpy - Shock to Sales



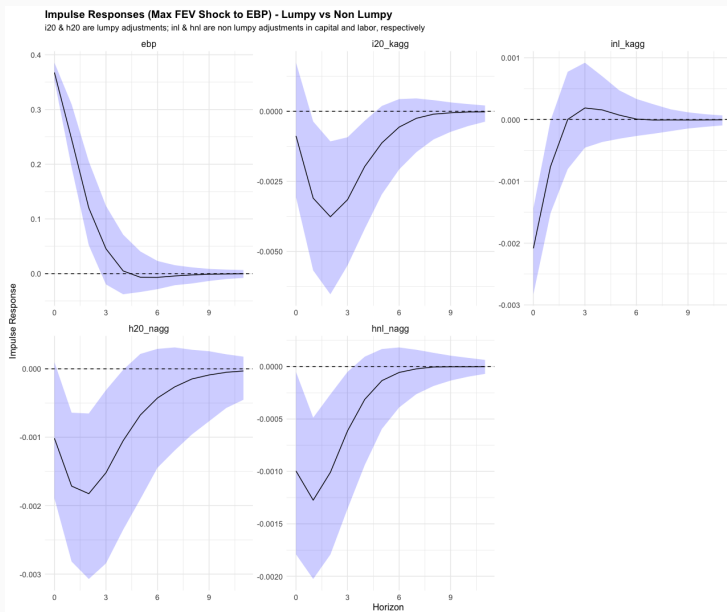
- Response in Lumpy (20) vs Non Lumpy Investment Rate (First Row), and
- Response in Lumpy (20) vs Non Lumpy Employment/Hiring Rate (Second Row).

Impulse Responses - Lumpy vs Non Lumpy - Shock to GDP



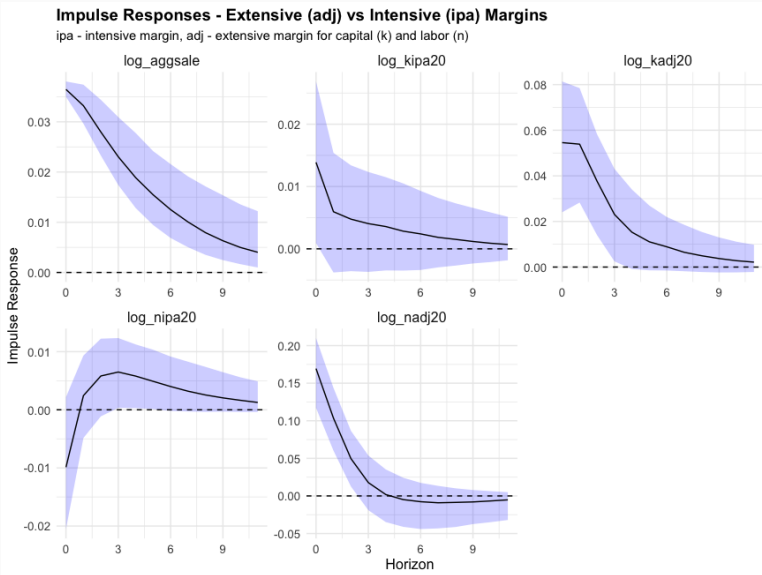
- Significant responses in both lumpy margins.
- Similar Responses when targeting GDP or (Aggregated) Firm Sales.

Impulse Responses - Lumpy vs Non Lumpy - Shock to Excess Bond Premium



- Significant Negative Cycles.

Impulse Responses - Intensive vs Extensive - Shock to Sales

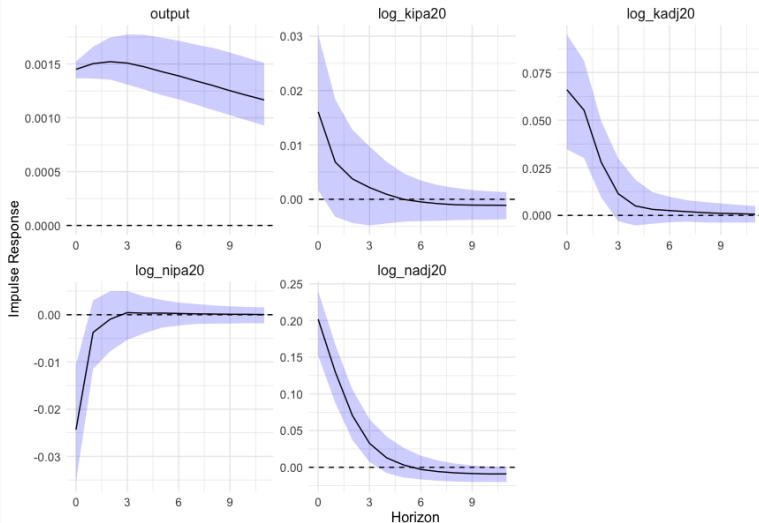


- Intensive (ipa) vs Extensive (adj) adjustments in capital (First Row) and Employment (Second Row).

Impulse Responses - Intensive vs Extensive - Shock to GDP

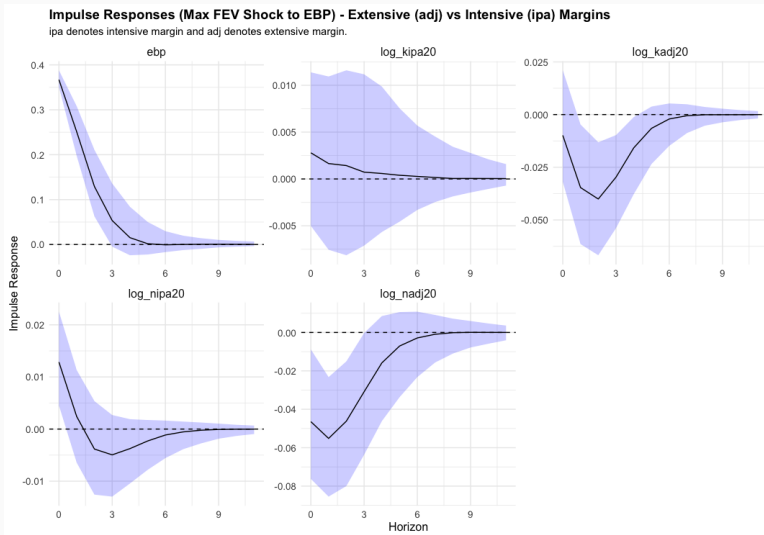
Impulse Responses (Max FEV Shock to GDP) - Extensive (adj) vs Intensive (ipa) Margins

ipa - intensive margin, adj - extensive margin for capital (k) and labor (n)



- Extensive margin is more important.

Impulse Responses - Intensive vs Extensive - Shock to Excess Bond Premium



- Similar pattern emerges: extensive margin drives the adjustment

Results - Firm Dynamics

- Two-way mapping: aggregate response $\Leftrightarrow$ firm level responses.
- Time varying loadings imply time varying impulse responses at the firm level.
- Can group responses impulse responses by firm characteristics:
 - **Size:** Small vs Big Size Firms.
 - **Debt Growth:** Low vs High Debt Growth Firms
 - **Sales Growth:** Low vs High Sales Growth Firms
 - **Intangible Intensity:** Low vs High Intangible Investment Firms
 - **Capital Age:** Newer vs Older Capital Vintage Firms - Age
- Type of Aggregate Shock - **Shock to EBP**

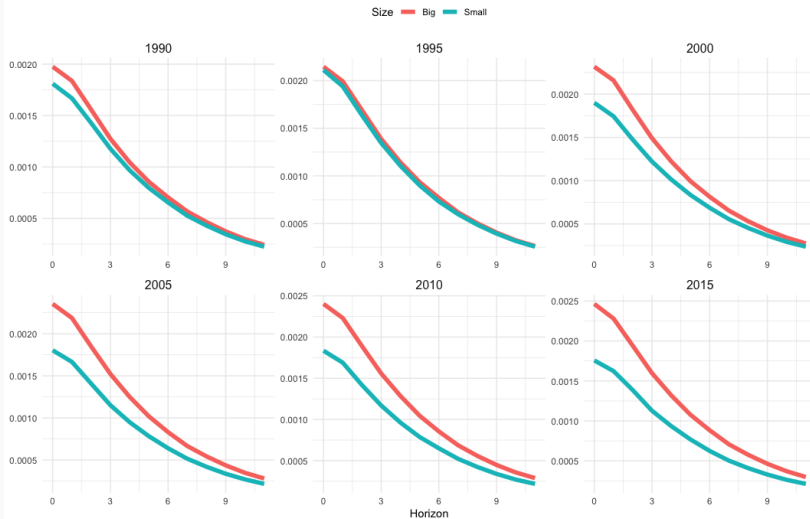
Firm-Level Responses: Summary

- We map firm-level investment responses to aggregate shocks using the DFM-implied investment rate ($x_{i,t}$). This allows a direct micro–macro linkage between aggregate impulses and heterogeneous firm behavior.
- Responses vary systematically across the firm distribution. Larger firms, high-growth firms, and firms with younger capital vintages exhibit stronger and more pronounced investment-rate responses.
- Financial shocks (EBP shocks) reveal substantial cross-sectional dispersion. Small (and financially constrained firms) display negative or muted investment responses, while large firms show minimal or no contraction, highlighting the role of financial frictions.
- Heterogeneity becomes even more pronounced when examining extreme (quantile-based) firm groupings:
 - **Size:** Very Small vs. Very Large Firms
 - **Sales Growth:** Very Low vs. Very High Sales-Growth Firms

(Median) Investment Rate Response at the Firm Level - By Size (I)

(Median) Firm Level Investment Rate Responses by - Firm Size

Small and Big Firms based on 20th and 80th quantiles of Total Assets of firms in that year.



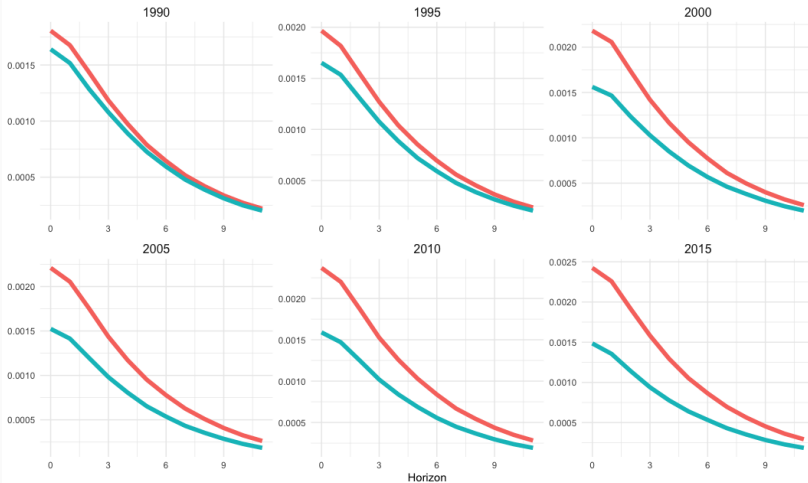
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(Median) Investment Rate Response at the Firm Level - By Size (II)

(Median) Firm Level Investment Rate Responses by - Firm Size

Small and Big Firms based on 10th and 90th quantiles of Total Assets of firms in that year.

Size — Big — Small

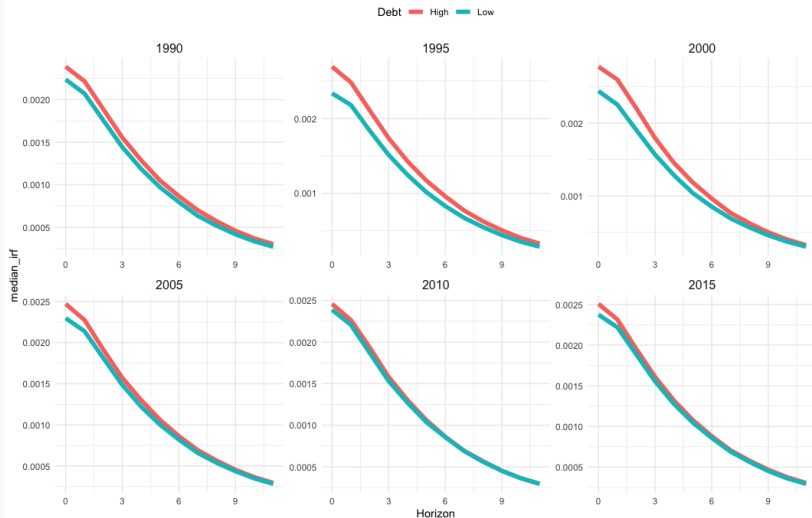


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Investment Rate Response at the Firm Level - By Debt Growth

(Median) Firm Level Investment Rate Responses for Firms by - Debt Growth

Low vs High debt growth firms based on the 20th and 80th quantiles of debt growth in that year.

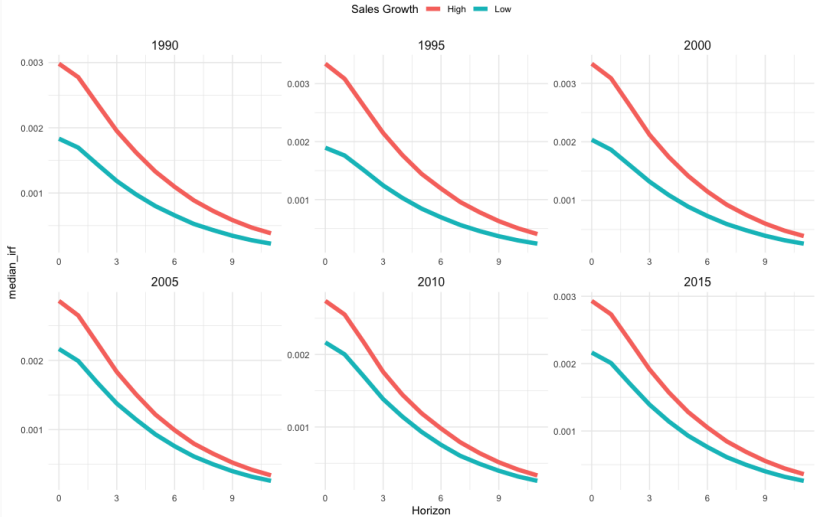


More pronounced effects as you go to more extreme quantiles. High debt growth firms have bigger investment rate responses. [Back](#)

Investment Rate Response at the Firm Level - By Sales Growth (I)

(Median) Firm Level Investment Rate Responses for Firms by - Sales Growth

Low vs High sales growth firms based on the 20th and 80th quantiles of sales growth in that year.

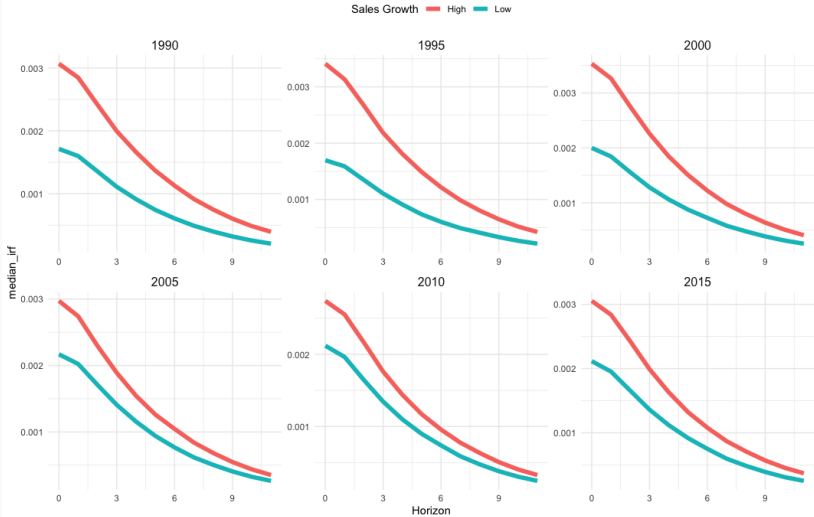


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Investment Rate Response at the Firm Level - By Sales Growth (II)

(Median) Firm Level Investment Rate Responses for Firms by - Sales Growth

Low vs High sales growth firms based on the 10th and 90th quantiles of sales growth in that year.

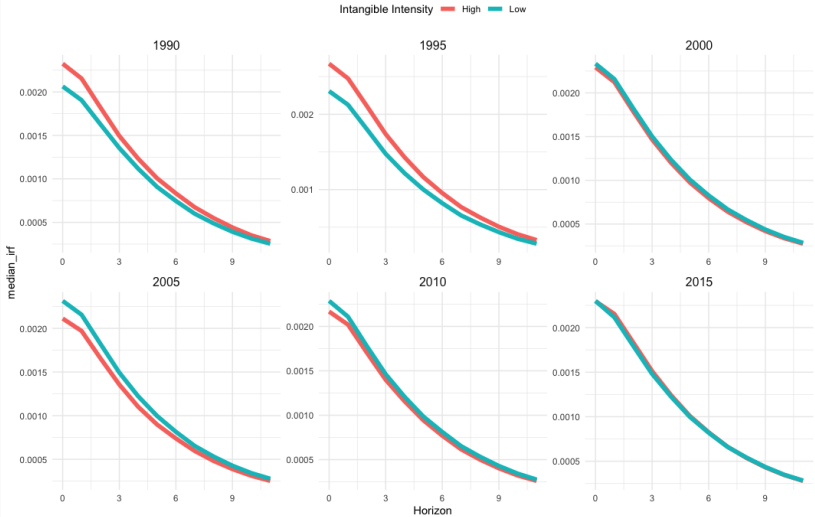


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Investment Rate Response at the Firm Level - By Intangible Intensity

(Median) Firm Level Investment Rate Responses for Firms by - Intangible Intensity

Low vs High Intangible Intensity firms based on the 20th and 80th quantiles of intangible intensity in that year.

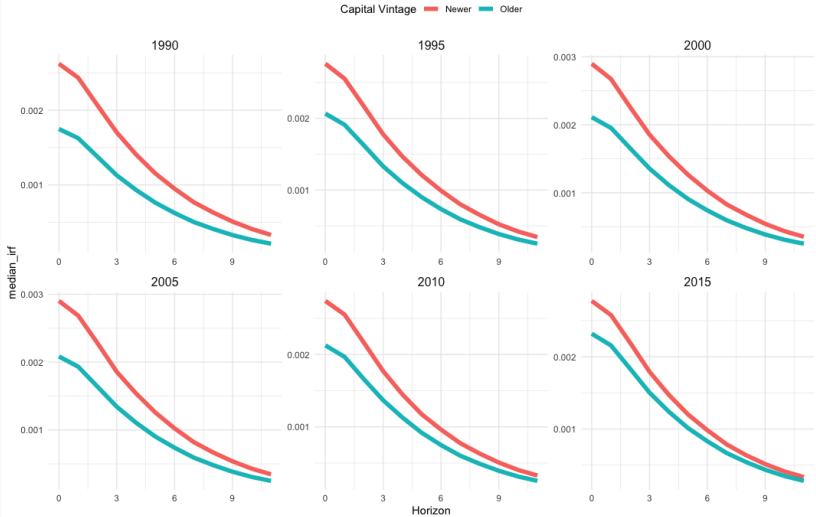


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Investment Rate Response at the Firm Level - By Installed Capital Age

(Median) Firm Level Investment Rate Responses for Firms by - Capital Age

Newer vs Older Capital Vintage firms based on the 20th and 80th quantiles of capital age in that year.

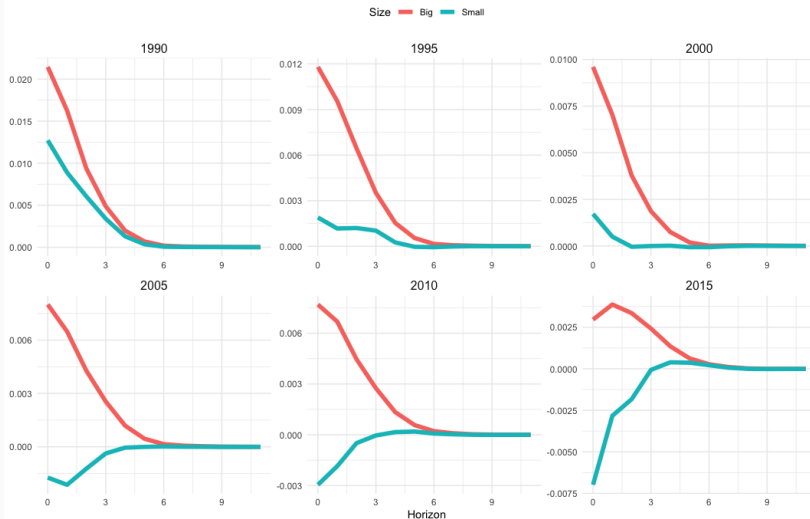


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Investment Rate Response at the Firm Level - By Firm Size - Aggregate EBP Shock

(Median) Firm Level Investment Rate Responses by - Firm Size - Shock to EBP

Small and Big Firms based on 20th and 80th quantiles of Total Assets of firms in that year.



Results from Khan Thomas (2008)

Simulated Data

- Heterogeneous plants with predetermined k , aggregate shock A , idiosyncratic productivity z , and a **stochastic fixed adjustment cost** ξ drawn each period (the key nonconvexity). Small “maintenance” investments avoid the fixed cost.
- Each plant compares the value of **adjusting to its target** k^* (paying ξ) vs. **not adjusting** (letting $(1 - \delta)k$ prevail). This yields a threshold rule: adjust iff $\xi \leq \xi^*(z, k)$.
- Lumpiness arises because depreciation pulls inactive plants away from their targets; occasional low- ξ draws trigger **large jumps** in capital. Idiosyncratic shocks spread adjustment timing across plants.
- Main contribution: a DSGE model consistent with micro evidence—spikes, asymmetry, limited inaction—while showing that in general equilibrium the aggregate impact of lumpiness is **small** due to price-induced smoothing.

Equilibrium Conditions

- Firms choose labor and capital (adjust or not) to maximize their value given p and the shock A .
- Representative household supplies labor inelastically and owns all firms; p is the marginal value of output in units of consumption.

Aggregate Firm Value

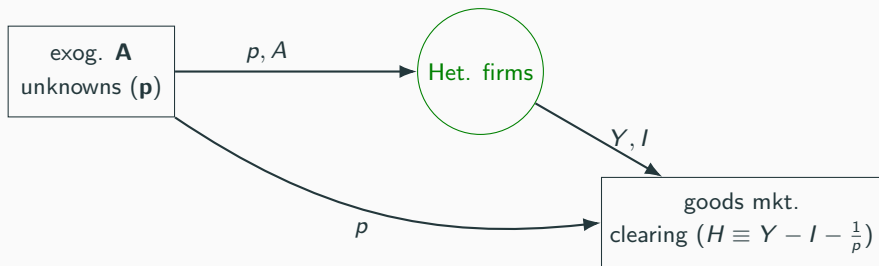
$$V(z, k) = -\phi \int_0^{\xi^*(z, k)} \xi dG(\xi) + G(\xi^*)V^A(z, k) + [1 - G(\xi^*)]V^{NA}(z, k)$$

Goods-Market Clearing (GE Condition)

$$\frac{1}{p} = \iint A z k^\alpha n(z, k, \xi)^\nu d\mu(z, k) dG(\xi) - \iint (k'(z, k, \xi) - (1 - \delta)k) d\mu(z, k) dG(\xi)$$

- $\mu(z, k)$ is the stationary distribution over firm states.
- **Equilibrium:** price p solves the system $H(p, A) = 0$ given firm policies and μ .

Model in DAG Form



The market equilibrium condition is

$$H(p, A) = 0$$

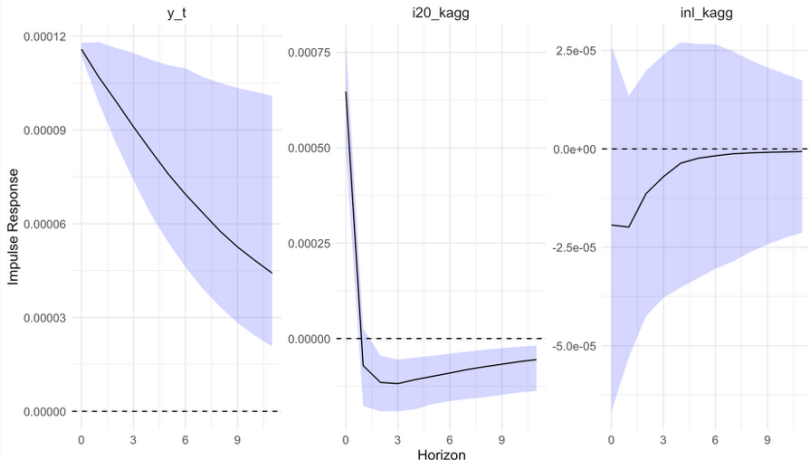
where $H(p, A) = Y - I - \frac{1}{p}$. This sequence formulation enables the solution of the model quickly using the Sequence Space Jacobian algorithm of Auclert et al (2021).

After solving the model, we simulate panel data for 8000 firms and a lot of years. DFM-FAVAR model is then estimated on this data (T=50 years) and impulse responses to aggregates are plotted following a business cycle shock.

Simulated Data - Lumpy vs Nonlumpy Responses

Impulse Responses on Model Simulated Data - Lumpy vs Non Lumpy

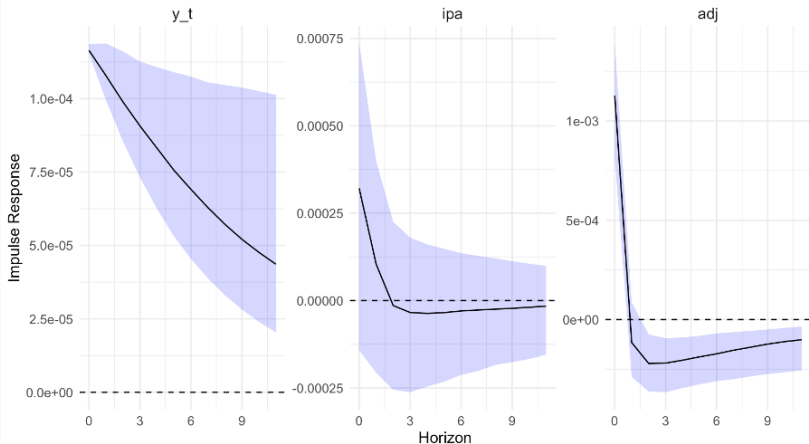
(Max FEV Shock to Aggregate Output); i20 - lumpy adjustments & inl - non lumpy adjustments in capital



Simulated Data - Extensive vs Intensive Margin Responses

Impulse Responses on Model Simulated Data - Extensive vs Intensive Margins

(Max FEV Shock to Aggregate Output); ipa - Intensive & adj - Extensive



Conclusion

- Using rich firm-level data, we find strong evidence that **aggregate business cycle movements are driven by lumpy adjustment dynamics at the firm level.**
- The **extensive margin**—the share of firms undertaking large capital changes—**plays a far more important role than the intensive margin** (the size of such changes).
- Both business cycle and financial shocks generate **heterogeneous firm-level responses** that vary systematically by size, leverage, and investment intensity.
- Lumpy adjustments in capital and labor frequently occur **synchronously** within firms, amplifying aggregate comovement.
- The findings carry important policy implications related to monetary policy, business dynamism and growth.